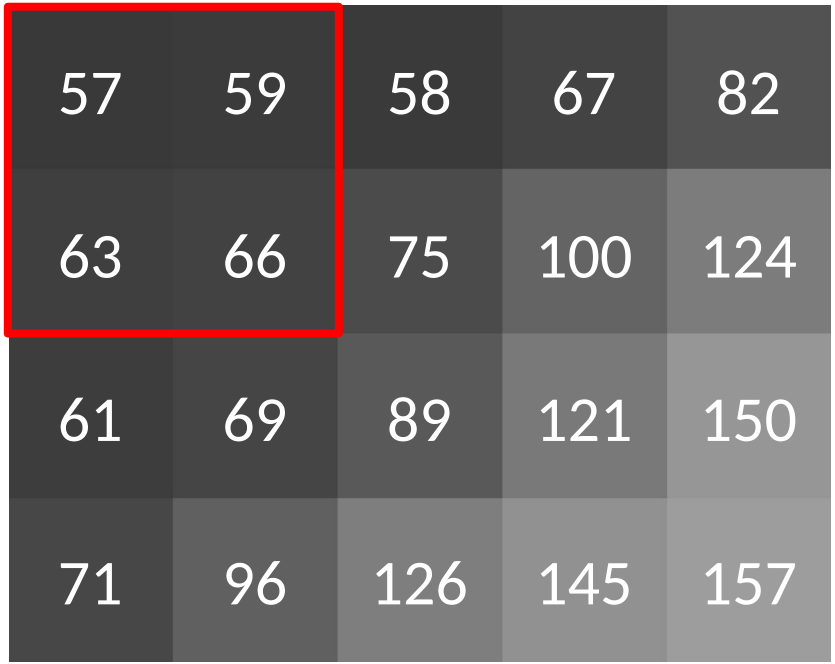
Computer Vision with Embedded Machine Learning

Pooling Layer

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Average Pooling

Image



Window: 2x2
Stride: 2
Find average under window:
(57 + 59 + 63 + 66) / 4 = 61.25

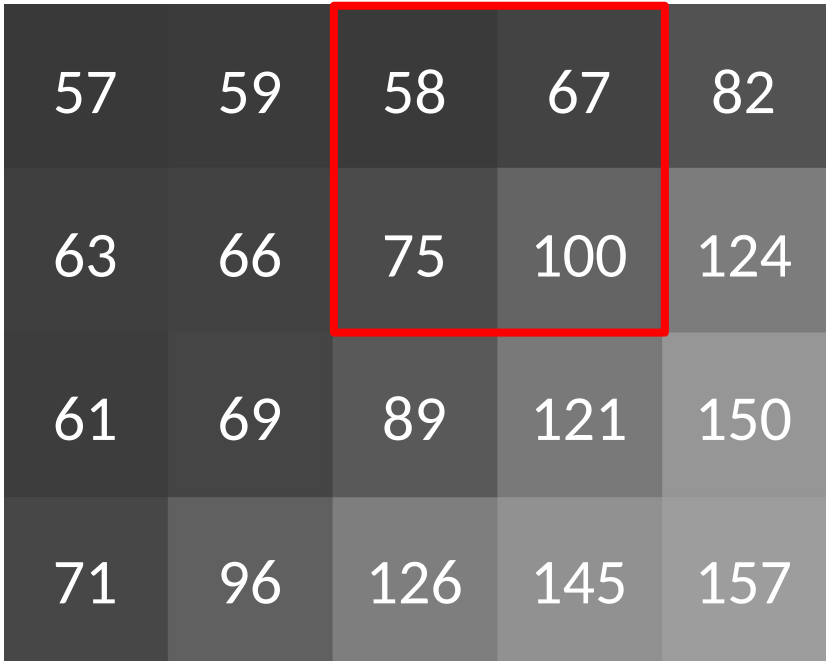
Output

61	

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Average Pooling

Image



Window: 2x2
Stride: 2
Find average under window:
(58 + 67 + 75 + 100) / 4 = 75

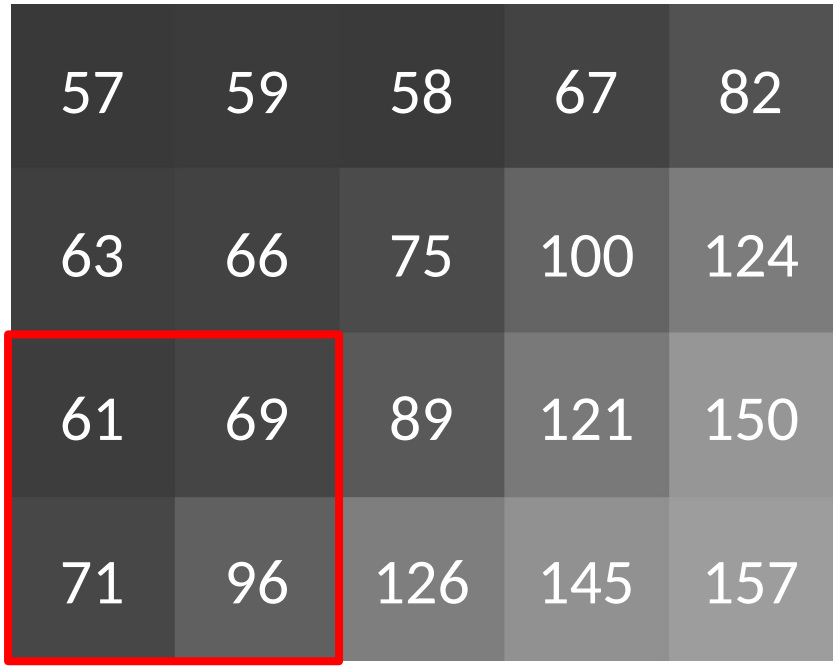
Output

61	75

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Average Pooling

Image



Window: 2x2
Stride: 2
Find average under window:
(61 + 69 + 71 + 96) / 4 = 74.25

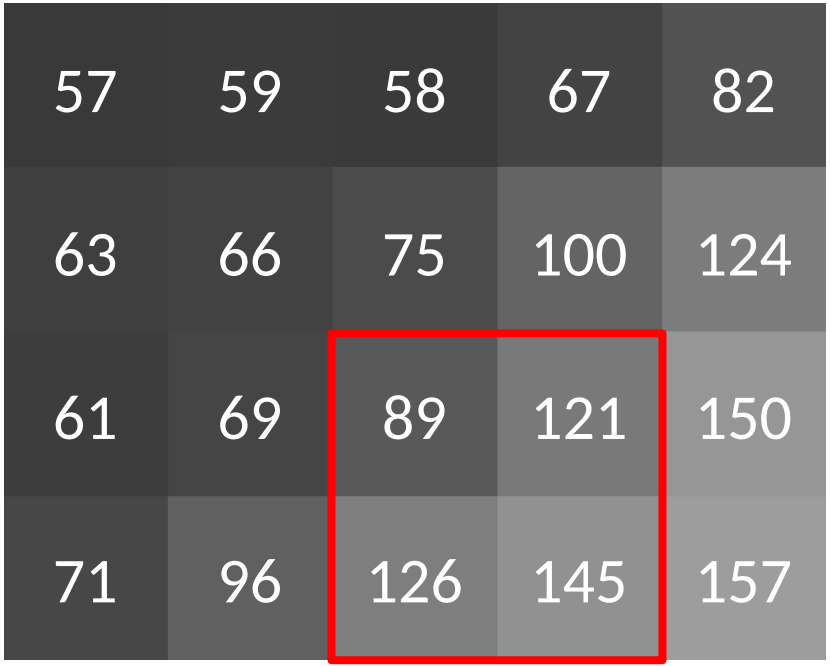
Output

61	75
74	

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Average Pooling

Image



Window: 2x2
Stride: 2
Find average under window:
(89 + 121 + 126 + 145) / 4 = 120.25

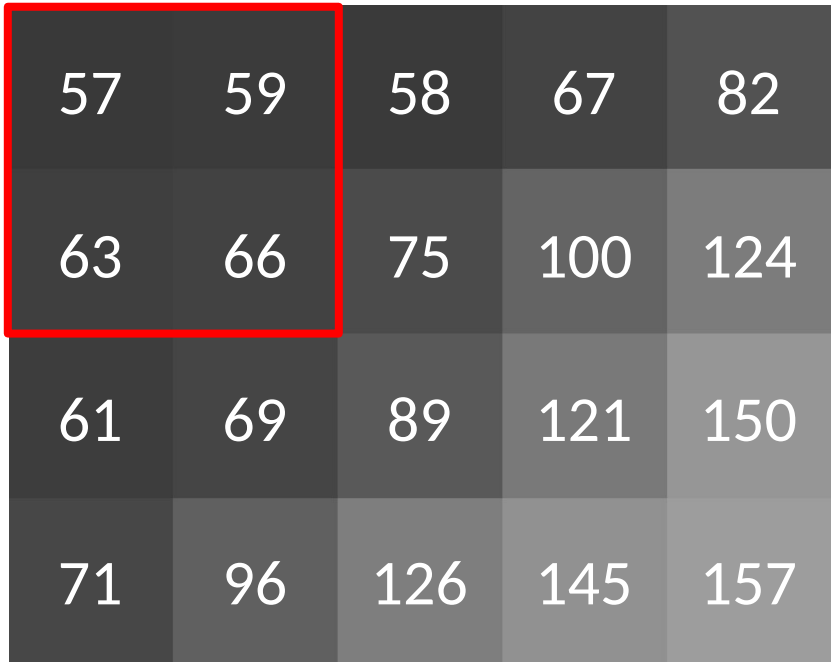
Output

61	75
74	120

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Max Pooling

Image



Window: 2x2
Stride: 2

Find maximum value under window: 66

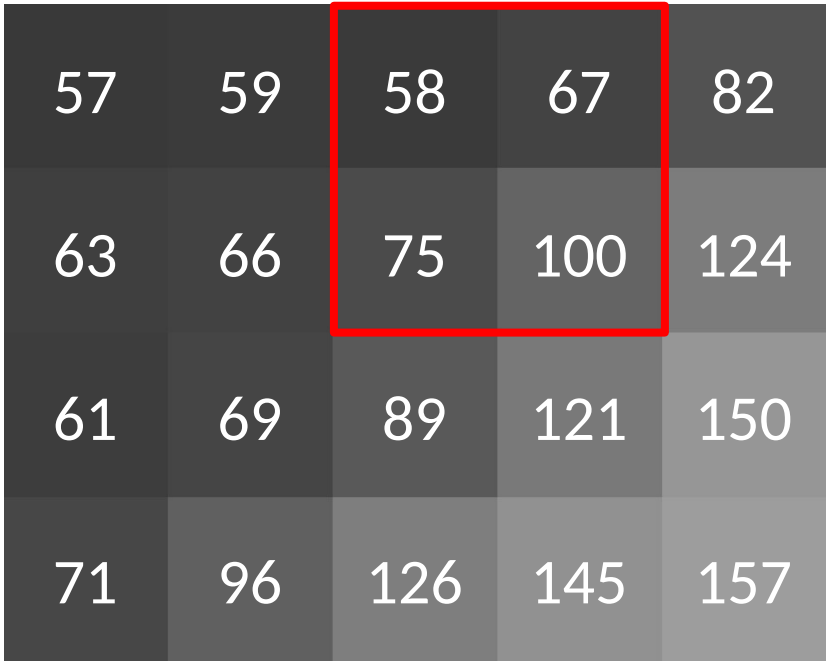
Output

66	

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Max Pooling

Image



Window: 2x2
Stride: 2

Find maximum value under window: 100

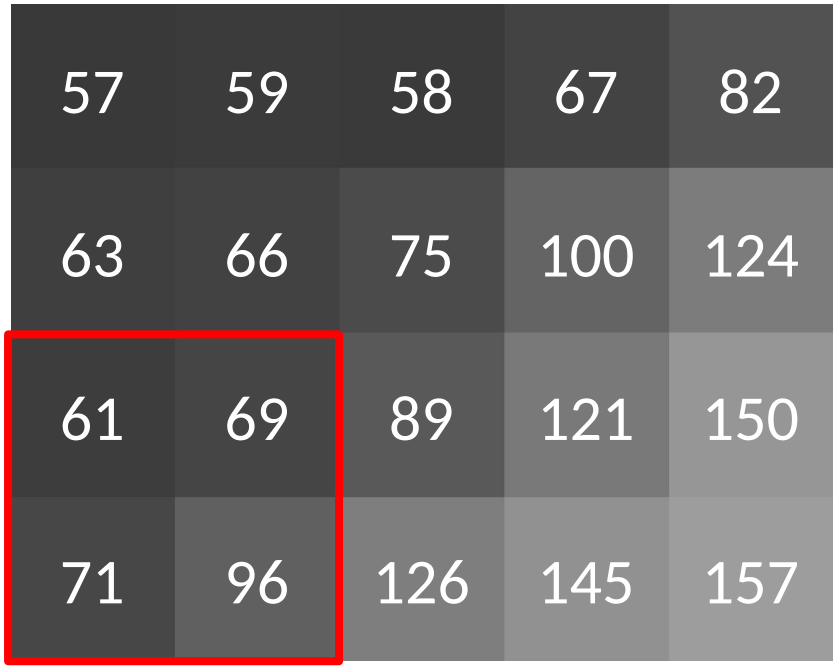
Output

66	100

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Max Pooling

Image



Window: 2x2
Stride: 2

Find maximum value under window: 96

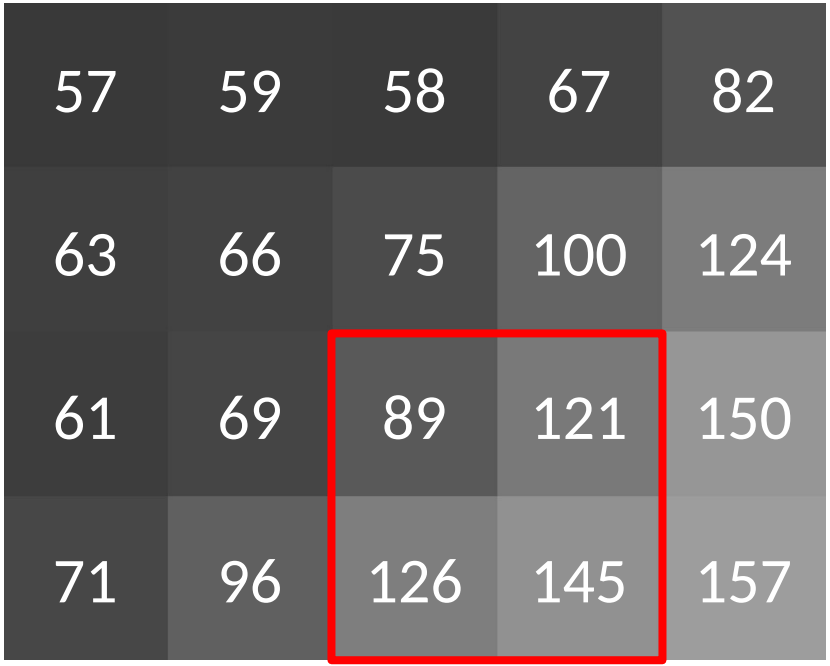
Output

66	100
96	

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Max Pooling

Image



Window: 2x2
Stride: 2

Find maximum value under window: 145

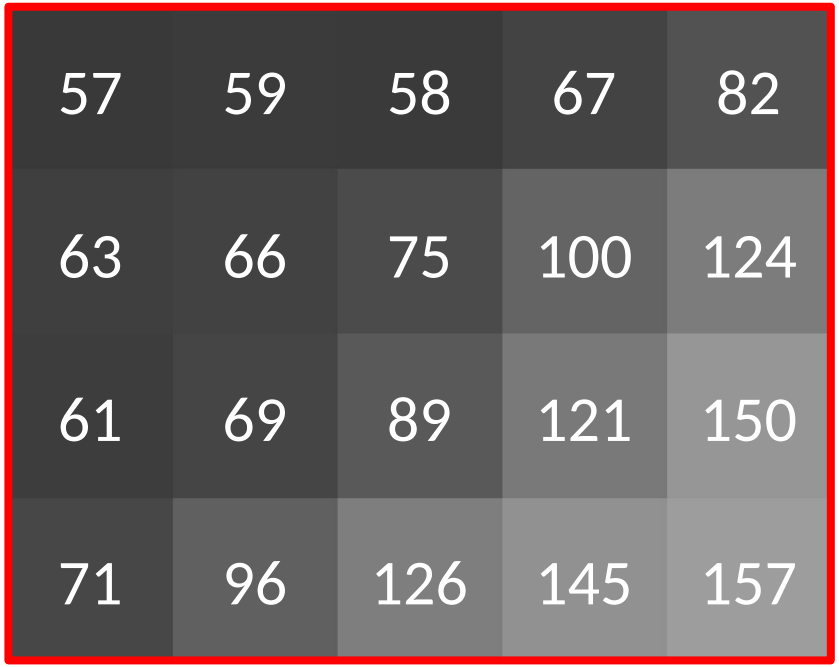
Output

66	100
96	145

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Global Pooling

Image



Global average output: 91.8

92

Global max output: 157

157

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Filtered image
(198x128)



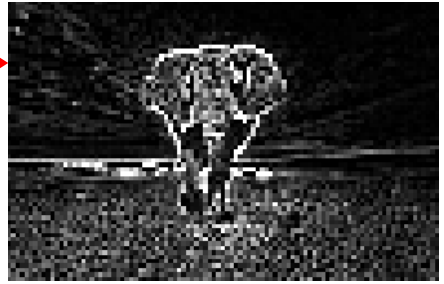
Average Pooling

(99x64)



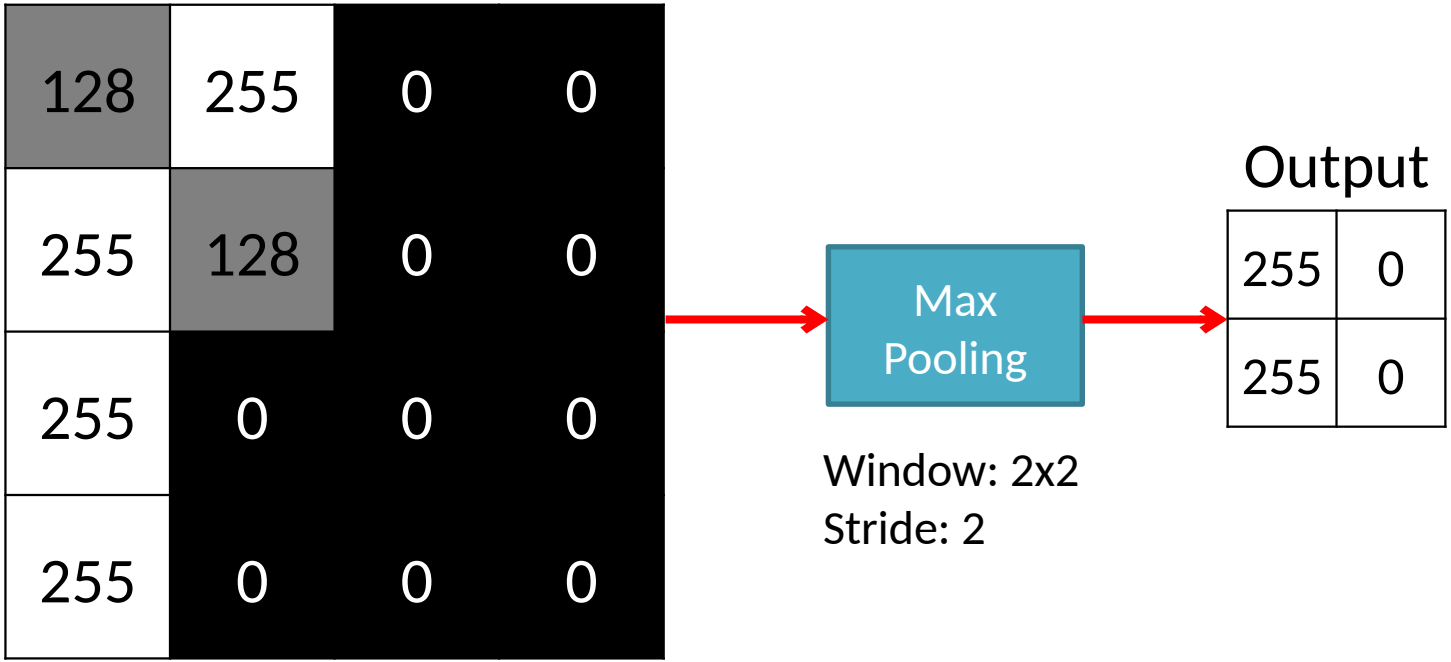
Max Pooling

(99x64)



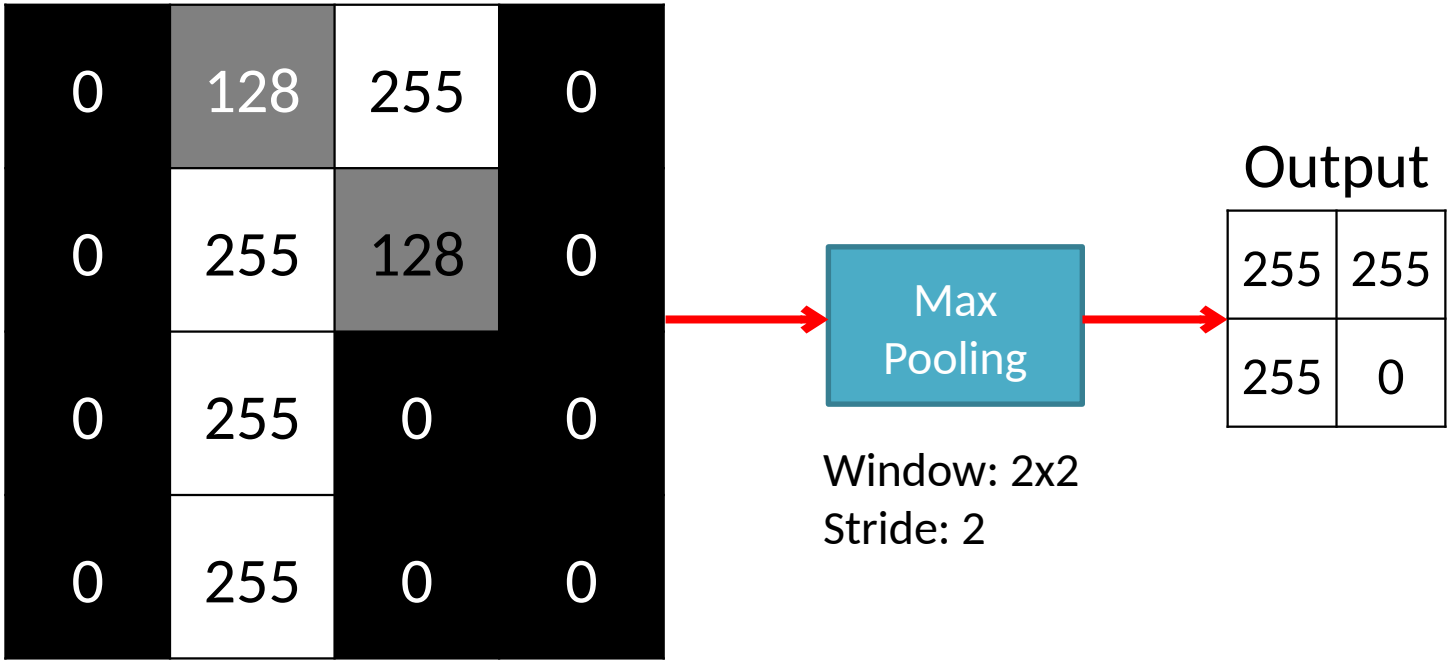
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Local Translation Invariance



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Local Translation Invariance



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